

Lesson 4: Parameters and Return Make

45 minutes

Overview

Using Programming Patterns and a step-by-step approach students make their own version of a Rock-Paper-Scissors app. At the beginning of the lesson students are able to explore a working version of the app but begin with a blank screen. Students use an Activity Guide to go through the high level steps they should use to develop their app but leaves it to them to decide how to write the code. At the end students submit their apps and answer a free-response question in the style of a Create performance task written response prompt. Student apps and written responses can be assessed using a provided rubric.

Standards

Full Course Alignment

CSP Conceptual Framework

- **CRD-2** - Developers create and innovate using an iterative design process that is user-focused, that incorporates implementation/feedback cycles, and that leaves ample room for experimentation and risk-taking.

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

Build the Rock Paper Scissors App

Wrap Up (5 minutes)

Assessment: Grading the Project

Objectives

Students will be able to:

- Debug programs that use functions with parameters and return
- Write functions with parameters and return values that meet a set of specified requirements

Preparation

- Review the app students will need to make and ideally spend some time working on the project yourself to anticipate the challenges students will encounter
- Review the Free-Response Supports and consider which, if any, supports are suitable for your classroom.
- Check the **"Teacher's Lounge"** forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 7 - Lesson 4 - Parameters and Return Make** - Slides
▼ Make a Copy
- **Writing Support - Parameters and Return Make** - Handout
▼ Make a Copy

- [CSP Parameters and Return Make - Rock Paper Scissors App](#) - Activity Guide [▼ Make a Copy](#)

Teaching Guide

Warm Up (5 minutes)

 **Discuss:** *How do parameters and return change the way you write programs?*

Have students share their ideas with a partner and then discuss as a class.

Discussion Goal: This discussion should prompt students to better understand what turns out to be a slightly different make project. Today they're going to be taking on "blank functions" rather than "blank screens". This is because with parameters and return values you can chunk out your programs into pieces that have well-defined behavior. Remind students that this is known as procedural abstraction. If students don't land on these ideas don't aim to force it but return to this idea in the wrap up.

Remarks

 Today we're going to be working on another Make project, but this one is a little bit different than others we've done in the course. A lot of the code will already be written for us, and we'll be working on blank functions, rather than entirely blank programs. Thanks to the way that parameters and return work, it's easy to design programs this way.

Activity (35 minutes)

Build the Rock Paper Scissors App

Teaching Tip

Supporting Students: While students are working on their apps, circulate the room and check in with students who need a little help. Encourage students to collaborate and discuss bugs with each other.

Debugging: Review with students steps they can use to debug if they get stuck:

- Build small parts of the program at a time and test them to make sure they work
- Run the code on turtle mode
- Add the variables to the watcher
- Explain the code to a friend and clarify the differences between what they expect to happen and what is actually happening in

Group: Make a determination as to whether this project will be completed in pairs or individually. You may even choose to let students decide.

 **Level 1 - Explore:** Have students explore the working Rock Paper Scissors app in Level 1. If students are not working in pairs they should still discuss with a neighbor.

- What does each button do
- How does the screen get updated after clicking each button

Review: Quickly review what students will be doing again before they move on to the next step.

Teaching Tip

Make lessons typically ask students to take on an entirely blank screen when writing their program. In this instance students have to take on the "blank function" which needs to fit within the larger program they're developing. This is part of a broader mindset shift that students develop in this unit. Functions with parameters and return values operate like their own "mini programs" which can be written and tested independently. There is actually less difference than students might think between previous make projects, in which students get the user interface elements and no code, and this project, in which they get some starter code but blank functions with parameters. In each case they need to develop programs that integrate into a larger portion of the project. The focus of this unit is how functions with parameters and return values help further split programs into logical chunks like this.

Distribute: Give students copies of **CSP Parameters and Return Make - Rock Paper Scissors App** if you will be using it during the class.

Do This: Direct students to level two where they complete the Rock Paper Scissors App. Based on the needs of your classroom decide whether you will collectively go through the activity guide or have students complete it individually. Afterwards give them time to work on their projects and circulate the room to offer support. Students who finish early can work on the extensions suggested in the activity guide.

Submit: Encourage students to check the rubric on the last page of the Activity Guide before submitting.

Teaching Tip

Adapting Make Lessons - You may decide that you'd like to have your students write the code for a modified or entirely different app using the same skills. This might help you differentiate your instruction or provide you with a way to reassess students. Check out the [r customizemakelessons_how-to-guide/csp/2025] for guidance on how to get started with this.

Wrap Up (5 minutes)

Remarks

Great work on your Rock, Paper, Scissors app today. Before wrapping up, we will practice writing for the style of questions that are asked as part of the Create performance task on the AP exam. Respond to the question in Step 5 in your activity guide, and submit it along with your project.

Do This: Complete the written response in Step 5 of the Activity Guide.

Teaching Tip

Each student, in every classroom, will come to the written response with a different level of comfort, ability, and familiarity with writing for these style of questions. As such, consider offering one or more of the following supports:

Writing Support: Provide students with these sentence starters for guiding their response.

App Interface Support: For students who did not fully complete their app, provide them with this snippet from the exemplar code.

 **Discuss:** *How could using parameters and return help you write programs collaboratively?*

Discussion Goal: This should help review ideas that students saw in this project, in particular that parameters and return help break programs into chunks. It should also help them see that this approach could be used to collaboratively design programs. If teammates agree in advance what each function should do they can go off and write them separately before coming back together to make the entire app work.

Remarks

Parameters and return help us chunk programs into pieces. Each important piece can be made into its own function which can be written and tested separately from the others. If you and your partner agree in advance how to split up the way a program is written and are clear on what each function needs to do, you could develop a program just like this make project.

Assessment: Grading the Project

Rubric: Use the rubric provided with the project to assess student projects.

Assessment Opportunity

Student responses can be used as a formative assessment to evaluate their understanding of connections between user interface features and functionality. Review their explanations for clarity, accuracy, and alignment with the AP exam's expectations. The provided rubric can be used to guide this review, and to help provide students with feedback to guide their learning and improvement.



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